

About This Document

The Pseudocode / Technical section tests fundamentals of data structures, algorithms, complexity and program output.

These concepts also recur in the technical interview for SP/DSE roles.

Data Structures and Complexity

12 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. What is the time complexity of binary search on a sorted array of n elements?

(A) $O(n)$

(B) $O(\log n)$ ✓

(C) $O(n \log n)$

(D) $O(1)$

Answer: (B) $O(\log n)$

Q2. Which data structure works on the FIFO principle?

(A) Stack

(B) Queue ✓

(C) Tree

(D) Graph

Answer: (B) Queue

Q3. Which data structure works on the LIFO principle?

(A) Queue

(B) Stack ✓

(C) Linked List

(D) Heap

Answer: (B) Stack

Q4. What is the worst-case time complexity of bubble sort?

(A) $O(n)$

(B) $O(n \log n)$

(C) $O(n^2)$ ✓

(D) $O(\log n)$

Answer: (C) $O(n^2)$

Q5. Which traversal of a binary search tree yields elements in sorted order?

(A) Preorder

(B) Postorder

(C) Inorder ✓

(D) Level order

Answer: (C) Inorder

Q6. The average-case time complexity of a lookup in a hash table is:

(A) $O(1)$ ✓

(B) $O(n)$

(C) $O(\log n)$

(D) $O(n \log n)$

Answer: (A) $O(1)$

Q7. A tree with n nodes has how many edges?

(A) n

(B) $n-1$ ✓

(C) $n+1$

(D) $2n$

Answer: (B) $n-1$

Q8. Which of the following is NOT a linear data structure?

(A) Array

(B) Stack

(C) Queue

(D) Tree ✓

Answer: (D) Tree

Q9. The recurrence $T(n) = 2T(n/2) + n$ solves to:

(A) $O(n)$

(B) $O(n \log n)$ ✓

(C) $O(n^2)$

(D) $O(\log n)$

Answer: (B) $O(n \log n)$

Q10. The maximum number of nodes at level L of a binary tree (root at level 0) is:

(A) $2L$

(B) L^2

(C) 2^L ✓

(D) $2^L - 1$

Answer: (C) 2^L

Q11. The space complexity of the standard merge sort is:

(A) $O(1)$

(B) $O(\log n)$

(C) $O(n)$ ✓

(D) $O(n^2)$

Answer: (C) $O(n)$

Q12. Which sorting algorithm is stable and has guaranteed $O(n \log n)$ time?

(A) Quick sort

(B) Merge sort ✓

(C) Selection sort

(D) Heap sort

Answer: (B) Merge sort

Output and Logic

6 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. `function f(n){ if(n<=1) return 1; return n*f(n-1); }` What does `f(4)` return?

(A) 12

(B) 24 ✓

(C) 16

(D) 6

Answer: (B) 24

Q2. `for(i=0; i<3; i++) print(i);` What is the output?

(A) 1 2 3

(B) 0 1 2 ✓

(C) 0 1 2 3

(D) 0 1

Answer: (B) 0 1 2

Q3. `x=10; y=20; x=x+y; y=x-y; x=x-y;` What are the final values of `x` and `y`?

(A) `x=10, y=20`

(B) `x=20, y=10` ✓

(C) `x=30, y=10`

(D) `x=20, y=20`

Answer: (B) `x=20, y=10`

Q4. `sum=0; for i in 1..n: for j in 1..n: sum=sum+1.` Final value of `sum` is:

(A) `n`

(B) `2n`

(C) `n^2` ✓

(D) `n log n`

Answer: (C) `n^2`

Q5. The postfix expression for `(A + B) * C` is:

(A) `AB+C*` ✓

(B) `ABC+*`

(C) `A+BC*`

(D) `AB*C+`

Answer: (A) `AB+C*`

Q6. `count=0; n=18; while(n>0){ count++; n=n/2; }` (integer division). Final value of `count`?

(A) 4

(B) 5 ✓

(C) 6

(D) 3

Answer: (B) 5



Pseudocode & Technical MCQs

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

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Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

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